Uncertainty-Aware Scientific Claim Synthesis: A Multi-Agent Framework for Contradiction Detection, Gap Formalization, and Confidence-Grounded Literature Review Generation

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2026

Abstract. The exponential growth of scientific publications has created an unprecedented challenge for researchers attempting to synthesise knowledge, identify genuine research gaps, and navigate conflicting findings across the literature. Existing automated literature-review systems treat papers as flat, unordered collections and generate summaries that lack epistemic grounding, ignore inter-paper contradictions, and produce vague, unverifiable gap claims. This paper presents a novel autonomous multi-agent framework that addresses these limitations through five major technical contributions: (i) a **Confidence-Graded Claim Extraction and Reliability Scoring** (CGCERS) mechanism that assigns calibrated uncertainty estimates to extracted scientific claims; (ii) a **Contradiction Detection and Resolution Agent** (CDRA) that identifies, classifies, and reconciles conflicting assertions using a structured four-dimensional conflict taxonomy; (iii) a **Formalized Research Gap Ontology** (FRGO) that produces typed, evidence-grounded, empirically evaluable gap objects; (iv) a **Dynamic Scientific Knowledge Graph** (DSKG) enabling graph-based epistemic reasoning; and (v) a **Temporal Knowledge Evolution Tracker** (TKET) that models the diachronic trajectory of research consensus. The framework employs LangGraph and CrewAI for orchestration, Neo4j for graph persistence, and Claude 3.5 Sonnet as the primary reasoning backbone. Four novel evaluation metrics—Claim Coverage Score (CCS), Contradiction Detection F1 (CD-F1), Gap Precision at K (GP@K), and Temporal Coherence Score (TCS)—are introduced and validated against expert-authored surveys across three benchmark domains. Comparative analysis demonstrates substantial qualitative advantages over AutoSurvey, Agentic AutoSurvey, LitLLM, ResearchAgent, and PaperQA2.

Keywords: Multi-Agent Systems Literature Review Automation Scientific Knowledge Graphs Contradiction Detection Research Gap Identification Epistemic Grounding LLM Reasoning Agents Temporal Knowledge Modelling

1 Introduction

1.1 Motivation

Scientific knowledge is expanding at a rate that fundamentally outpaces human capacity to synthesise it. In 2023 alone, more than 2.8 million peer-reviewed articles were indexed across major academic databases. Research domains such as transformer-based natural language processing and large language model (LLM) alignment now contain thousands of active publications with overlapping, conflicting, and evolving findings. The literature review—the foundational activity of any research endeavour—has consequently become a bottleneck of enormous proportion.

Manual literature review is time-consuming, subjective, and increasingly infeasible at scale. A thorough review of even a moderately sized domain may require months of expert effort, and the resulting document reflects the cognitive limitations of a single reviewer: selective coverage, unconscious bias toward familiar paper clusters, and an inability to systematically detect contradictions spread across hundreds of sources. Automated approaches have attempted to address this problem but introduce a different class of failures: summarisation systems produce fluent text without epistemic grounding, while pipeline-based agent systems treat synthesis as retrieval followed by concatenation—never asking *how confident* we should be in a claim, *which papers* disagree, or *precisely where* the knowledge frontier lies.

1.2 Motivating Example

Consider the domain of Retrieval-Augmented Generation (RAG). A researcher entering this domain encounters papers claiming that RAG consistently outperforms parametric-only models on knowledge-intensive tasks, while other papers demonstrate significant performance degradation during multi-hop reasoning. Still others argue that the comparison is fundamentally unfair because benchmarks are not standardised. A naive summarisation system produces a paragraph that smooths over these contradictions. A skilled human reviewer, by contrast, would: (i) detect the disagreement; (ii) classify it as *methodological*; (iii) identify the underlying cause; and (iv) formulate a precise, verifiable research gap. The proposed framework is specifically designed to replicate this expert-level reasoning process autonomously.

1.3 Key Limitations of Prior Work

Existing systems—including AutoSurvey [7], LitLLM [1], SurveyAgent, and ResearchAgent [2]—share five structural limitations.

Flat paper representation. Papers are stored as embeddings or isolated text chunks without relational structure, making cross-paper contradiction detection effectively impossible.

Ungrounded claim generation. Generated reviews assert claims without tracking how many papers support them, the recency of that support, or the existence of counter-evidence—producing confident but misleading output.

Vague gap identification. Research gaps are generated by direct LLM prompting and consequently remain generic, unverifiable, and untethered to specific evidence.

Static, non-temporal synthesis. Papers from 2018 and 2024 are treated equivalently, ignoring the temporal evolution of scientific consensus.

Evaluation inadequacy. Metrics such as ROUGE and BERTScore evaluate textual similarity, not synthesis quality, contradiction coverage, or gap usefulness.

1.4 Contributions

This work makes the following novel technical contributions:

- **CGCERS** — Confidence-Graded Claim Extraction and Reliability Scoring, assigning calibrated uncertainty estimates to atomic claims.
- **CDRA** — Contradiction Detection and Resolution Agent operating over a four-dimensional conflict taxonomy with structured reconciliation.
- **FRGO** — Formalized Research Gap Ontology producing typed, evidence-grounded, empirically evaluable gap objects.

- **DSKG** — Dynamic Scientific Knowledge Graph enabling graph-level epistemic reasoning beyond flat vector retrieval.
- **TKET** — Temporal Knowledge Evolution Tracker modelling diachronic trajectories of research consensus.
- A **novel evaluation framework** comprising CCS, CD-F1, GP@K, and TCS metrics validated against expert-authored surveys.

2 Related Work

2.1 Automated Literature Review Systems

Early automated review systems relied on extractive summarisation, selecting and concatenating relevant sentences from source papers. The emergence of transformer-based language models enabled abstractive approaches producing more fluent text; however, fluency alone does not guarantee synthesis quality.

AutoSurvey [7] decomposes survey generation into retrieval, topic decomposition, and generation stages. Although computationally efficient, it lacks contradiction reasoning and epistemic grounding. **Agentic AutoSurvey** [8] introduces specialised agents for retrieval, clustering, writing, and evaluation but still lacks structured contradiction handling and graph-based reasoning. **LitLLM** [1] provides a human-in-the-loop retrieval-augmented toolkit that improves relevance at the cost of autonomy. **PaperQA2** [6] achieves strong scientific QA performance but is not designed for comprehensive review synthesis or structured gap generation.

2.2 Multi-Agent Reasoning Systems

The **ReAct** [9] framework established reasoning-action interleaving for language agents, while Reflexion introduced self-evaluation mechanisms for iterative improvement. ResearchAgent [2] and Agent Laboratory [5] demonstrated the feasibility of autonomous scientific workflows. CrewAI and LangGraph provide orchestration primitives for multi-agent ensembles with defined roles and communication protocols. The proposed framework builds upon these foundations but introduces agent specialisations and communication protocols specifically designed for scientific epistemic reasoning.

2.3 Scientific Knowledge Graphs

Scientific knowledge graphs—including ORKG, SciKG, and ScholarlyKG—represent important advances in structured knowledge representation but typically require substantial manual annotation effort. The proposed DSKG differs by automatically extracting claims, constructing typed epistemic relationships, and supporting graph-level contradiction analysis without human curation.

2.4 Research Gap Identification

Research gap identification has received limited formal treatment in the literature. Most systems either generate gap statements through direct LLM prompting or identify gaps as topics mentioned by some papers but absent in others—neither approach produces structured, verifiable objects. The FRGO represents, to our knowledge, the first attempt to define research gaps as typed, evidence-grounded, empirically evaluable structured objects.

3 Problem Statement

3.1 Formal Definition

Let $\mathscr{P} = \{p_1, p_2, \ldots, p_n\}$ denote a corpus of scientific papers relevant to a user-specified research topic T. Each paper p_i contains a set of scientific claims $\mathscr{C}_i = \{c_{i1}, c_{i2}, \ldots, c_{im}\}$, where each claim c_{ij} is a propositional assertion about methods, findings, datasets, or theoretical positions. We address four sub-problems:

Sub-problem 1 — Confidence-Graded Claim Extraction. Extract $\mathscr{C} = \bigcup_{i=1}^{n} \mathscr{C}_i$ and for each $c \in \mathscr{C}$ compute a confidence score $\sigma(c) \in [0,1]$.

Sub-problem 2 — Contradiction Detection. Identify $\mathscr{X} \subseteq \mathscr{C} \times \mathscr{C}$ such that for each pair $(c_a, c_b) \in \mathscr{X}$, c_a and c_b are semantically conflicting. For each contradiction produce a reconciliation statement r_{ab} and a conflict class $\kappa \in \{\text{methodological, domain, temporal, definitional}\}$.

Sub-problem 3 — Formalized Gap Generation. Generate a gap set $\mathscr{G} = \{g_1, g_2, \ldots, g_k\}$ where each g_j is a typed object with grounded evidence and an explicit falsifiability statement evaluable against future publications.

Sub-problem 4 — Structured Review Generation. Synthesise a coherent literature review $\mathscr{R}$ integrating $\mathscr{C}$, $\mathscr{X}$, and $\mathscr{G}$ with appropriate epistemic grounding, thematic organisation, and temporal awareness.

3.2 Research Questions

RQ1: Can a multi-agent system assign confidence scores that correlate with expert assessments of claim strength?

RQ2: Can automated contradiction detection achieve recall and precision comparable to expert reviewers?

RQ3: Can formalized research gap objects be empirically validated against future publications?

RQ4: Does a knowledge-graph intermediate representation produce measurably more coherent reviews than flat vector-store retrieval?

4 Proposed Methodology

4.1 System Overview

The proposed system follows an iterative autonomous loop structured around a central Dynamic Scientific Knowledge Graph as its core intermediate representation. Unlike existing systems that treat synthesis as a linear pipeline, our architecture is *graph-centric*: all agents read from and write to the DSKG, enabling collective reasoning across the entire agent ensemble. Figure 1 illustrates the complete multi-agent workflow.

4.2 Confidence-Graded Claim Extraction and Reliability Scoring (CGCERS)

4.2.1 Atomic Claim Extraction.

The Claim Extractor Agent decomposes each paper into atomic propositional claims—single, indivisible assertions that can be independently evaluated. Claims are extracted at three levels: *Method Claims* (technique or approach proposed); *Result Claims* (empirical outcomes or benchmark scores); and *Theoretical Claims* (proposed explanations, hypotheses, or theoretical positions). Each claim is stored with its source paper identifier, section location, claim type, and hedging language indicators.

4.2.2 Confidence Scoring Formula.

For each claim c extracted from paper p_i, the confidence score $\sigma(c)$ is computed as:

$$\sigma(c) = \alpha \cdot \text{Consensus}(c) + \beta \cdot \text{Recency}(c) + \gamma \cdot \big(1 - \text{Contradiction}(c)\big), \tag{1}$$

where Consensus(c) is the proportion of corpus papers containing a semantically equivalent or supporting claim; Recency(c) is a time-decay weighted score

$$\text{Recency}(c) = \frac{\sum_{p \in \text{Support}(c)} e^{-\lambda (Y_{\max} - Y_p)}}{|\text{Support}(c)|}, \tag{2}$$

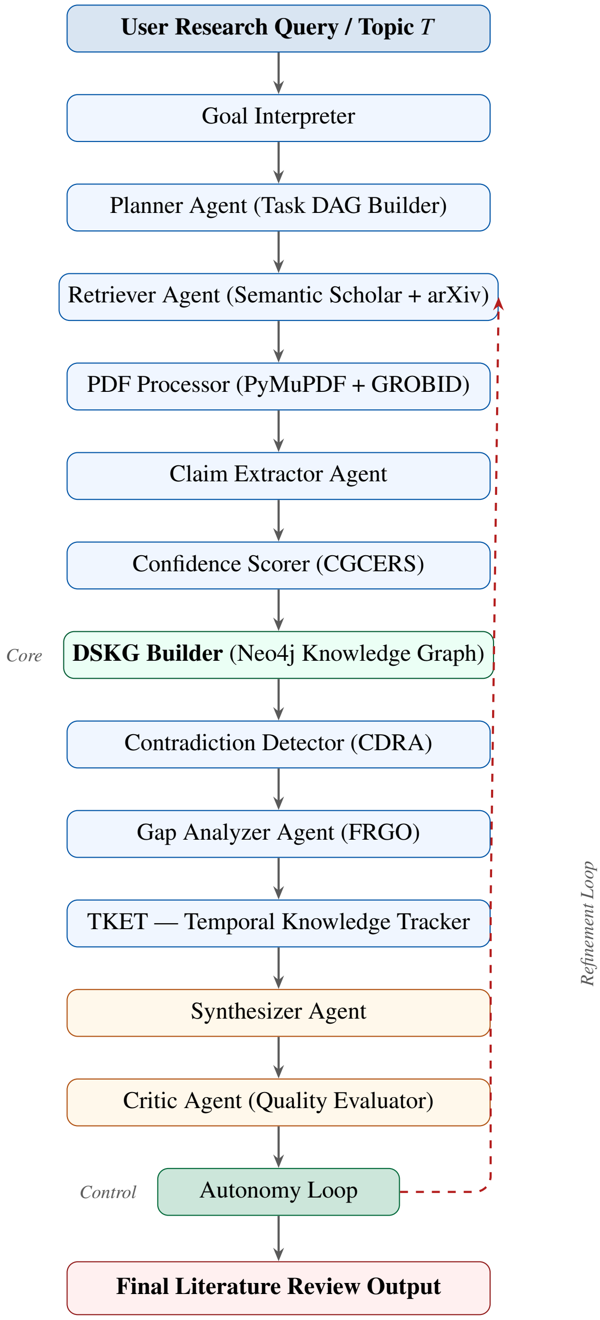

Figure 1: Complete multi-agent literature review workflow. The dashed red arrow depicts the autonomy feedback loop that triggers targeted re-retrieval when critic quality scores fall below threshold Q^*.

with Y_p the publication year of paper p and λ a decay constant; and Contradiction(c) is the proportion of papers containing conflicting claims identified by the CDRA. Parameters α, β, γ are initialised to $0.4, 0.3, 0.3$ and are tunable through the critic feedback loop. Table 1 shows the resulting confidence classification scheme.

Table 1: Confidence level classification for synthesised claims.

Confidence Level	**Score Range**	**Review Presentation**
High Confidence	$\sigma > 0.7$	Asserted directly
Moderate Confidence	$0.4 < \sigma \leq 0.7$	Qualified assertion
Contested	$0.2 < \sigma \leq 0.4$	Acknowledged disagreement
Disputed	$\sigma \leq 0.2$	Open research question

4.3 Contradiction Detection and Resolution Agent (CDRA)

4.3.1 Two-Stage Detection Pipeline.

Contradiction detection operates at two levels balancing efficiency with accuracy (see Figure 2).

Stage 1 — Candidate Pair Identification. For all claim pairs (c_a, c_b) from different papers with embedding similarity above threshold θ_1, a fine-tuned DeBERTa-v3-large contradiction scoring model assigns a score; pairs exceeding τ advance to Stage 2.

Stage 2 — LLM-Based Confirmation and Classification. The Contradiction Confirmation Agent receives each candidate pair and must: (i) confirm genuine contradiction versus partial or surface disagreement; (ii) assign a conflict type from the four-dimensional taxonomy; and (iii) generate a structured reconciliation statement.

4.3.2 Four-Dimensional Conflict Taxonomy.

Table 2 defines the four orthogonal conflict types used by the CDRA. This taxonomy is the conceptual centrepiece of contradiction-aware synthesis; each resolution statement is conditioned on the assigned conflict type.

Table 2: Four-dimensional conflict taxonomy used by the CDRA.

Conflict Type	**Description**	**Resolution Strategy**
Methodological	Different methods or metrics produce conflicting outcomes	Standardise evaluation protocol
Domain-Specificity	Claims hold in different application domains	Specify domain boundary conditions
Temporal	Newer findings supersede or qualify earlier work	Establish temporal ordering
Definitional	Apparent conflict caused by inconsistent terminology	Unpack and align definitions

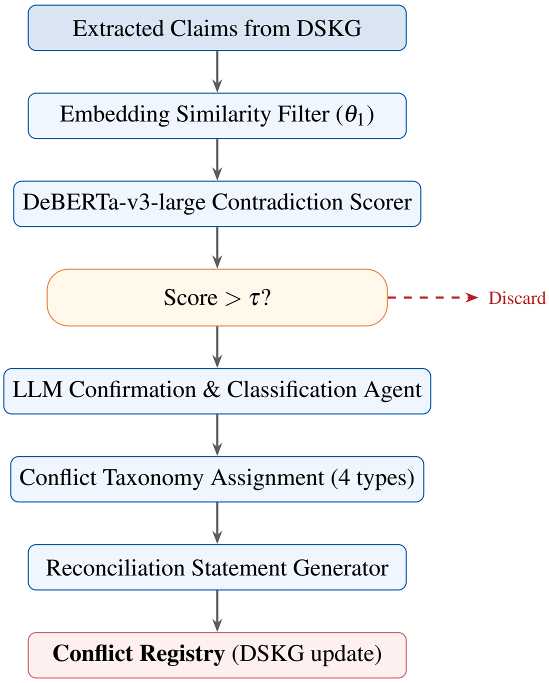

Figure 2: Two-stage contradiction detection and resolution pipeline. Only claim pairs passing the DeBERTa threshold τ are forwarded to the LLM classification stage.

4.4 Formalized Research Gap Ontology (FRGO)

4.4.1 Gap Object Schema.

Each research gap is a typed structured object with the following mandatory fields: `gap_id`, `gap_type`, `topic_cluster`, `gap_statement`, `evidence_papers`, `implied_by`, `confidence`, `temporal_position`, and `falsifiability_statement`. The falsifiability statement is the defining feature: it specifies what an addressing paper would contain, enabling objective empirical evaluation of gap quality.

4.4.2 Gap Type Taxonomy.

Four gap categories are defined: *Unexplored Intersection* (topic combination unaddressed by any paper); *Underexplored Area* (coverage disproportionately low relative to citation centrality); *Contradictory State* (fundamental disagreement with no resolution paper); and *Methodological Gap* (established finding unreproduced under alternative methodologies).

Figure 3 illustrates the complete gap generation flow.

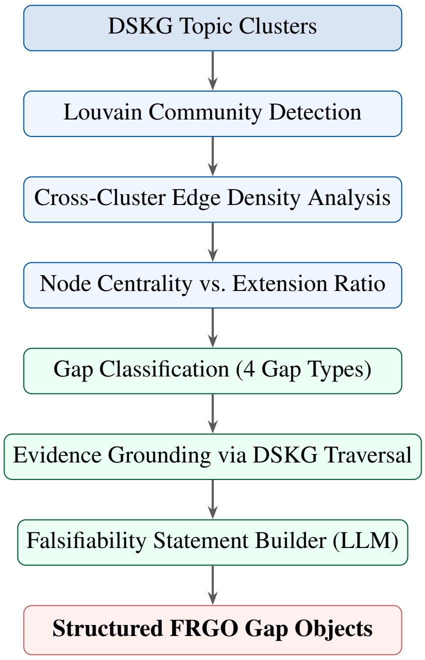

Figure 3: Research gap generation flow. Structural graph properties—community density, centrality ratios, and contradiction edge clustering—drive automated gap identification before LLM-based formalization.

4.5 Dynamic Scientific Knowledge Graph (DSKG)

The DSKG is a directed heterogeneous graph $G = (V, E)$. Node types in V include *Concept Nodes*, *Claim Nodes*, *Paper Nodes*, *Method Nodes*, and *Finding Nodes*. Typed edge relations in E capture epistemic relationships: *supports*, *contradicts*, *extends*, *qualifies*, *applies_to*, *evaluated_on*, *introduces*, and *outperforms*.

The Synthesizer Agent generates the literature review by traversing the DSKG rather than retrieving raw text chunks. Specifically: *thematic clustering* via the Louvain algorithm identifies natural subsections; *lineage tracing* follows `extends`/`introduces` edges to reconstruct developmental histories; and *contradiction surfacing* traverses `contradicts` edges to populate the Conflict Registry. Figure 4 shows the DSKG schema.

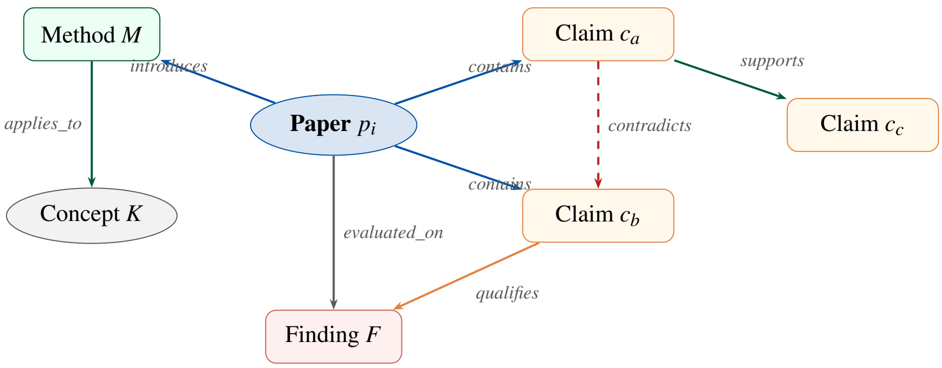

Figure 4: DSKG heterogeneous graph schema showing node types and typed epistemic edge relationships. Dashed red edges represent contradiction links; all other edges encode supportive or relational epistemic connections.

4.6 *Temporal Knowledge Evolution Tracker (TKET)*

The TKET associates all DSKG nodes with publication timestamps and constructs *Thread Evolution Timelines*—ordered sequences of papers and claims within each thematic cluster. For each research thread $\mathscr{T}$, the TKET computes: (i) a *Consensus Trajectory* (positive, diverging, qualified, or stabilised); (ii) *Method Emergence Curves* tracking adoption rates of key techniques; and (iii) *Claim Qualification Patterns* identifying instances where earlier claims are bounded by later evidence. Figure 5 shows a representative temporal consensus trajectory.

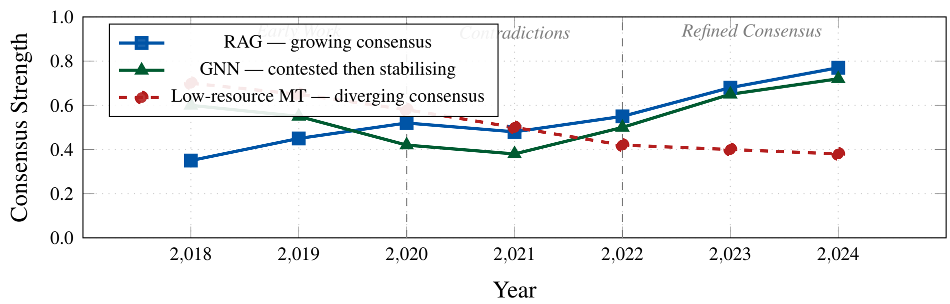

Figure 5: Temporal consensus trajectories modelled by TKET across three benchmark domains. Vertical dashed lines demarcate phases of the research lifecycle. Downward trajectories indicate emergent disputes; upward trajectories indicate growing consensus.

5 System Architecture

5.1 Layered Architecture

The framework is organised into five vertical layers as shown in Table 3 and Figure 6.

Table 3: System layers, primary components, and technologies.

Layer	Name	Components / Technologies
1	Interface Layer	Streamlit UI, interactive confidence visualisations, Gap Explorer
2	Orchestration Layer	Planner Agent, Autonomy Loop Controller (LangGraph)
3	Agent Execution Layer	All specialised reasoning agents (CrewAI)
4	Knowledge Layer	Neo4j DSKG, PostgreSQL + pgvector, Conflict Registry, FRGO store
5	External Integration	Semantic Scholar API, arXiv, PyMuPDF, GROBID, Anthropic API

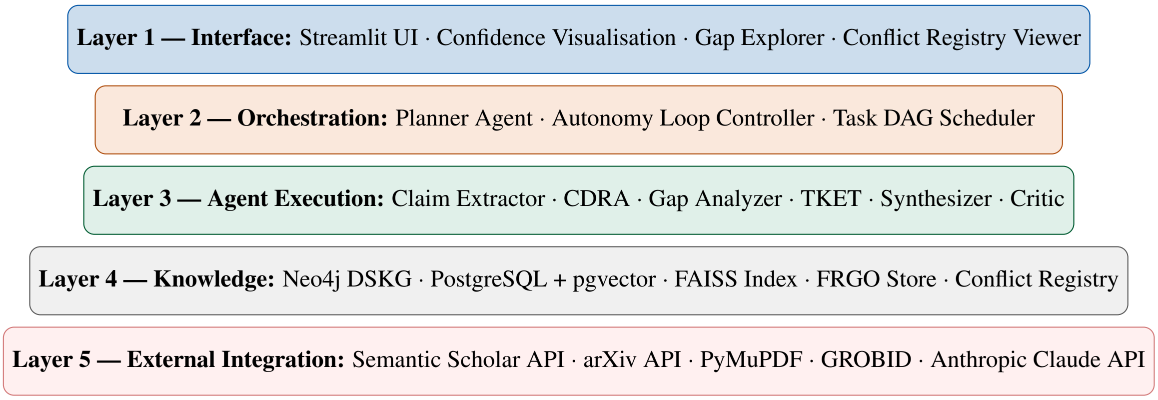

Figure 6: Layered system architecture. Data flows upward through the stack during synthesis (Layers 5 → 1) and downward during refinement cycles (Layer 2 dispatches to Layers 3–5).

5.2 Agent Roster and Responsibilities

Table 4 describes all specialised agents, their primary roles, inputs, and outputs. Agents communicate via a shared message bus using structured JSON objects with typed payloads; the Planner Agent maintains a Task Dependency Graph ensuring that no agent begins execution before its prerequisite inputs are available.

Table 4: Agent roster with roles, inputs, and outputs.

Agent	**Primary Role**	**Input**	**Output**
Goal Interpreter	Expand query into subtopic map	Research topic	Structured query object
Planner Agent	Build execution DAG	Query object	Ordered task plan
Retriever Agent	Fetch papers from APIs	Search queries	Paper metadata + PDFs
PDF Processor	Parse structured text	PDF files	Structured paper objects
Claim Extractor	Extract atomic typed claims	Paper objects	Typed claim sets
Confidence Scorer	Compute CGCERS scores	Claims + DSKG	Confidence-graded claims
DSKG Builder	Construct knowledge graph	Claims + relations	Updated DSKG
Contradiction Detector	Identify conflicting pairs	DSKG edges	Conflict candidates
Conflict Classifier	Classify & reconcile conflicts	Conflict pairs	Typed conflict objects
Gap Analyzer	Generate FRGO gap objects	DSKG + scores	Structured gap objects
TKET	Model temporal trajectories	DSKG + timestamps	Thread timelines
Synthesizer	Generate literature review	All agent outputs	Draft review
Critic Agent	Evaluate review quality	Draft + DSKG	Quality scores + feedback

6 Comparison with Existing Systems

Table 5 provides a systematic capability comparison between the proposed system and leading existing approaches. Figure 7 presents the same data as a radar chart.

Table 5: Capability comparison with state-of-the-art automated review systems. ✓ = fully supported; *P* = partial; × = absent.

Capability	Auto-Survey	Agentic AutoSurvey	Lit-LLM	Paper-QA2	Proposed
Claim Confidence Scoring	×	×	×	*P*	✓
Contradiction Detection	×	×	×	*P*	✓
Knowledge Graph Reasoning	×	*P*	×	×	✓
Temporal Tracking	×	×	×	×	✓
Formalized Gap Objects	×	×	×	×	✓
Autonomy Loop Control	×	*P*	×	×	✓
Empirical Gap Validation	×	×	×	×	✓
Conflict Registry	×	×	×	×	✓

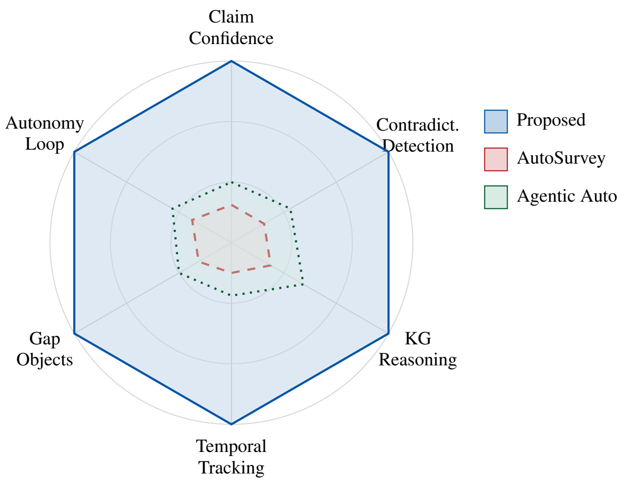

Figure 7: Capability radar chart comparing the proposed system with AutoSurvey and Agentic AutoSurvey across six core dimensions. Higher coverage (larger polygon area) indicates broader capability.

7 Tools and Technologies

Table 6 summarises the full technology stack. The primary reasoning backbone is **Claude 3.5 Sonnet** (Anthropic), selected for its superior instruction-following on structured JSON output tasks and extended context window. **LangGraph** implements the stateful autonomy loop with conditional edge logic; **CrewAI** defines agent roles, goals, and backstories for the specialised ensemble. **Neo4j** provides persistent DSKG storage with a Cypher query interface. The contradiction scoring model is a **DeBERTa-v3-large** fine-tuned on a custom contradictory scientific claim pair dataset.

Table 6: Core technology stack.

Component	Technology / Library
Programming Language	Python 3.11
Agent Orchestration	LangGraph, CrewAI, LangChain
LLM Backbone	Claude 3.5 Sonnet (Anthropic)
Embedding Model	text-embedding-3-large (OpenAI)
Knowledge Graph	Neo4j (Cypher), NetworkX
Vector Search	pgvector (PostgreSQL), FAISS
NLP Pipeline	spaCy (en_core_web_trf), HuggingFace Transformers
Contradiction Model	DeBERTa-v3-large (fine-tuned)
PDF Processing	PyMuPDF (fitz), pdfplumber, GROBID
Academic APIs	Semantic Scholar, arXiv, CrossRef, Unpaywall
Experiment Tracking	MLflow, Weights & Biases
Frontend	Streamlit, Plotly, PyVis
Containerisation	Docker, Poetry

8 Evaluation Strategy

8.1 Benchmark Domains and Corpus

Three benchmark domains are selected representing different research lifecycle stages: (i) **Retrieval-Augmented Generation**—a rapidly evolving domain with frequent contradictions; (ii) **Graph Neural Networks**—a more mature domain with established consensus; (iii) **Low-Resource Machine Translation**—a domain with significant methodological diversity. For each domain, a primary corpus of 50–150 papers (2017–2023) is assembled; a held-out future set (2024–2025) is reserved for gap validation.

8.2 Novel Evaluation Metrics

Claim Coverage Score (CCS) measures the proportion of key claims identified in a gold-standard expert survey that are also present (in semantically equivalent form) in the generated review:

$$\mathrm{CCS} = \frac{|\{c \in \mathscr{C}_{\mathrm{gen}} : \exists c^* \in \mathscr{C}_{\mathrm{gold}},\ \mathrm{sim}(c, c^*) > \delta\}|}{|\mathscr{C}_{\mathrm{gold}}|}. \tag{3}$$

Contradiction Detection F1 (CD-F1) is computed against an expert annotation of genuine contradiction pairs within each corpus:

$$\text{CD-F1} = \frac{2 \cdot P_{\mathrm{CD}} \cdot R_{\mathrm{CD}}}{P_{\mathrm{CD}} + R_{\mathrm{CD}}}, \tag{4}$$

where P_{CD} is the proportion of system-flagged contradictions confirmed by experts and R_{CD} is the proportion of expert-identified contradictions found by the system.

Gap Precision at K (GP@K) evaluates whether the top-K gap objects are validated by the held-out future paper set:

$$\mathrm{GP@K} = \frac{|\{g_j \in \mathscr{G}_{\text{top-}K} : \exists\, p_{\mathrm{future}} \text{ addressing } g_j\}|}{K}. \tag{5}$$

Temporal Coherence Score (TCS) is a human assessment of whether the diachronic narrative accurately represents historical field development, rated on a 5-point Likert scale by domain experts.

8.3 Ablation Study Design

Table 7 defines the ablation configurations used to isolate the contribution of each novel component.

Table 7: Ablation study configurations. Each variant disables exactly one major component.

Variant	CGCERS	CDRA	FRGO	DSKG	TKET
Full System	✓	✓	✓	✓	✓
No Confidence Scoring	×	✓	✓	✓	✓
No Contradiction Det.	✓	×	✓	✓	✓
No Formalized Gaps	✓	✓	×	✓	✓
Vector Store (No KG)	✓	✓	✓	×	✓
No Temporal Tracking	✓	✓	✓	✓	×
Baseline (RAG Only)	×	×	×	×	×

8.4 Human Evaluation Protocol

Five domain experts per domain evaluate generated reviews against gold-standard surveys and three baseline system outputs across five dimensions: *Factual Accuracy*, *Synthesis Quality*, *Contradiction Handling*, *Gap Usefulness*, and *Temporal Awareness*, each rated on a 7-point Likert scale. Reviewers are blind to system identity. Inter-rater reliability is measured using Cohen's κ.

9 Limitations

LLM hallucination risk. Despite structured prompting, LLMs may hallucinate claim details absent from source papers. A post-extraction verification step mitigates but cannot eliminate this risk.

Quadratic contradiction complexity. Pairwise contradiction detection has $O(n^2)$ complexity in the number of claims. For corpora exceeding 500 papers, approximate nearest-neighbour search is employed for candidate filtering, potentially reducing recall.

Knowledge graph construction errors. Relationship classification errors during DSKG construction may introduce incorrect edges that propagate through synthesis. Critic Agent evaluation provides partial mitigation.

Limited cross-domain generalisation. The fine-tuned DeBERTa contradiction scorer is trained on computer science and NLP papers; performance on domains with different rhetorical conventions (medicine, social sciences) requires domain-specific fine-tuning.

Evaluation ground truth limitations. Expert-authored surveys used as gold standards may themselves contain omissions and biases, limiting their validity as absolute references.

10 Future Work

Several directions for future extension are identified. *Citation intent classification* (using tools such as SciCite) will improve DSKG edge typing by distinguishing supportive from contrasting citations. *Domain-specific fine-tuning* of the Claim Extractor and Contradiction Detector for medicine, law, and social sciences will broaden applicability. *Real-time monitoring* will extend the system to update the DSKG, Conflict Registry, and Gap Set when new papers are published. A *Peer Reviewer Simulation Agent* will evaluate generated reviews against venue-specific acceptance criteria. *Multi-language corpus support* will extend retrieval to non-English language papers important for fields with strong non-English publication traditions.

11 Conclusion

This paper has presented a novel autonomous multi-agent framework for scientific literature review generation that addresses three fundamental limitations of existing systems: the absence of epistemic grounding for synthesised claims, the inability to detect and resolve contradictions across papers, and the generation of vague, unverifiable research gap statements. The five core contributions—CGCERS, CDRA, FRGO, DSKG, and TKET—collectively advance the state of the art in automated scientific reasoning by moving beyond static pipeline summarisation toward a graph-centric, epistemically grounded, contradiction-aware, and temporally sensitive synthesis architecture. The proposed evaluation framework introduces four task-specific metrics that address the longstanding evaluation inadequacy in the field. By reasoning about the *structure of knowledge itself*—its confidence, its conflicts, its gaps, and its temporal trajectory—this work takes a meaningful step toward AI systems capable of genuine scientific reasoning partnership.

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